

BENGALURU NORTH UNIVERSITY

Tamaka, Kolar – 563 103.



Curriculum, Credits Structure and Syllabus under SEP
for Final Year Single Major Under-Graduate
Programme of
Bachelor of Computer Applications (General), 2026-27

BOS IN COMPUTER APPLICATIONS (UG)
BENGALURU NORTH UNIVERSITY



BENGALURU NORTH UNIVERSITY
KOLAR - 563103

State Education Policy - 2024
(Semester Scheme)

Curriculum Structure for Bachelor of Computer Applications
(BCA)

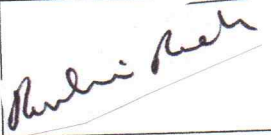
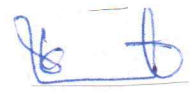
Syllabus for 5th and 6th Semesters

With Effect From

Academic Year 2026 - 2027 and onwards

Proceedings of the Board of Studies (BOS) in Computer Applications (UG)

The meeting of the Board of Studies (BOS) in Bachelor of Computer Applications (General) was held on 07/07/2026 at 10AM, Government First Grade College, K R Puram, Bengaluru in hybrid mode to consider and approve the syllabus for the 5th and 6th semesters of BCA (General) under Bengaluru North University. The Chairperson welcomed the members and presented the proposed syllabus. After discussion and deliberation, the Board unanimously approved the syllabus for the 5th and 6th semesters for implementation from the academic year 2026–27. The following BOS members were present:

Sl.No.	Name and Address	Designation	Signature
1	Dr. Murugan K, Associate Professor Department of Computer Science Government First Grade College K R Puram, Bengaluru-560036.	Chairperson	
2	Smt. Rashmi Rao K, Associate Professor Department of Computer Science Government First Grade College Hoskote, Bengaluru (Rural)	Member	
3	Dr. Sumanth S, Professor Department of Computer Science Government College for Women Kolar – 563101	Member	
4	Smt. Radhika A, Associate Professor Department of Computer Science Government First Grade College Chikkaballapura -562101	Member	
5	Dr. Hamela K, Associate Professor Department of Computer Science Government First Grade College Kolar-563101.	Member	
6	Sri. Manikandan S, Associate Professor Department of Computer Science Government College for Women Kolar – 563101	Member	
7	Dr. K. S. Manjunatha, Professor Department of Computer Science, Maharanis Science College for Women Mysore.	Member (External)	

Minutes:

1. The BOS members have approved the course structure and recommended (Theory and Practical).
2. The BOS members unanimously approved the proposed course structure for the newly introduced subjects.
3. The BOS members are accepted the detailed syllabus for the 5th and 6th semesters.

**Regulations, Scheme of study and Examination for BCA Degree Course under
Choice Based Credit System - Semester System (SEP Scheme)
(With effect from 2026 -2027)**

R1. a) Title of the course: **Bachelor of Computer Applications (BCA)**

b) Duration of the Course: Durations of the undergraduate programme shall extend SIX semesters (Three academic years) for the regular Bachelor Degree.

c) Scheme of study: there shall be six theory papers and two practical from first semester to five semesters.

d) There will be four theory, one practical and one project/Internship in sixth semester. The project work shall be carried out either independently or jointly (maximum of three students)

e) Medium of Instruction: The medium of instruction shall be English.

f) Scheme of Examination: At the end of each semester there is University Examination of three hours duration in each of the theory paper/practical.

R2. Each semester shall be of 90 working days from the date of commencement of the each Semester.

R3. Attendance: As per Bengaluru North University regulations in force for science degree courses.

R4. A Candidate is allowed to carry over all the previous uncleared (failed) theory papers/Practical to subsequent semesters as per Bengaluru North University regulations in force for science degree courses.

R5. The maximum period for completion of the course shall be as per Bengaluru North University regulations in force for science degree courses.

R6. Eligibility for admission:

a) A candidate who has passed the two years Pre-University Examination conducted by the Pre-University Education Board in Karnataka a minimum of 35% of marks.

b) A candidate who has passed Three years Diploma in Engineering of Government of Karnataka or any other examination considered as equivalent thereto shall be eligible for admission with minimum of 35% of marks in aggregate in all the semester /years.

c) Any student who has passed PUC –II Science, Arts or Commerce other than Karnataka securing with a minimum of 35% of marks.

R8. The total number of students to be admitted to the course shall be decided by the Bengaluru North University.

R9. Results: Results of candidate shall be declared and the classes awarded as per the procedure followed by the Bengaluru North University.

R10. Power to Remove Difficulties:

a) In case any difficulty arises in giving effect to the provisions of these regulations, the Vice-Chancellor shall order / make such provisions not inconsistent with the Act, Statutes, Ordinances or other Regulations, as appears to be necessary to expedient to remove the difficulty.

b) Every order made under this shall be subject to rectification by the appropriate University Authorities.

R11. The question paper pattern for theory paper has three sections. (80 Marks)

1. Section A includes 12 questions, out of which students shall attend 10 questions. Each question carries 2 Marks ($10 * 2 = 20$)
2. Section B includes 8 questions (question may contain sub questions), out of which students shall attend 6 questions. Each question carries 5 Marks ($6 * 5 = 30$)
3. Section C includes 4 questions (question may contain sub questions), out of which students shall attend 3 questions. Each question carries 10 Marks ($3 * 10 = 30$)

R12. Internal Assessment Scheme for Theory (4 Credits).

Assessment Criteria	Marks
Two test (Each carries 5 marks)	10
Assignment	5
Seminar	5
Total	20

R13. Internal Assessment Scheme for Theory (2 Credits).

Assessment Criteria	Marks
Test	5
Assignment	5
Total	10

R14. Internal Assessment Scheme for Practical (2 Credits).

Assessment Criteria	Marks
Test	5
Record	5
Total	10

R15. Evaluation Scheme for Practical Examination (SEE).

Assessment Criteria	Marks
Write up two programs (one from Section-A and Section-B)	20
Execution and output	15
Viva Voice based on Lab Activities	5
Total	40

Course Content for Bachelor of Computer Applications (BCA)

Curriculum Structure							
Program: BCA				Subject: Computer Applications			
Semester	Course Code	Title of the Paper	Credits (L+P)	No. of Teaching Hours / Per Week	Marks		Total Credits
					SEE	IA	
V	CA5T1	Web Technologies	4+0	4	80	20	20
	CA5T2	Data Mining	4+0	4	80	20	
	CA5T3	Software Engineering	4+0	4	80	20	
	CA5P1	Web Technologies Lab	0+2	4	40	10	
	CA5P2	Data Mining Lab	0+2	4	40	10	
	SEC2	Data Analytics Lab	0+2	4	40	10	
	CA5P3	Internship	0+2	4	40	10	
VI	CA6T1	PHP and MySQL	4+0	4	80	20	22
	CA6T2	Mobile Application Development	4+0	4	80	20	
	CA6T3	Deep Learning	4+0	4	80	20	
	CA6P1	PHP and MySQL Lab	0+2	4	40	10	
	CA6P2	Mobile Application Development Lab	0+2	4	40	10	
	DSS2	Cyber Security / Data and Information Security	2+0	2	40	10	
	CA6P3	Project Work	0+4	8	80	20	

Semester: V

CA5T1: Web Technologies

Course Code: CA5T1	Course Title: Web Technologies
Course Credits: 04	Teaching Hours/Week: 04
Total Contact Hours: 60	Exam Duration : 3 Hours
Internal Assessment : 20 Marks	Semester End Exam: 80 Marks

Course Outcomes (COs): After the successful completion of the course, the student will be able to:

CO1: Explain the fundamental concepts of the WWW, web technologies, communication protocols and the use of text editors.

CO2: Develop well-structured web pages using HTML by applying text formatting,

CO3: Design interactive web pages using JavaScript by implementing control structures, functions and arrays.

CO4: Design the web pages using CSS by applying selectors, the box model, positioning, tables, lists, transitions, transformations, and animations.

Course Content	Hours
Unit-I	15
Fundamentals of Web: Internet, WWW, Web Browsers, Web Servers, Webpages, Front End, Back End, Website, Static and Dynamic Websites. Overview of Protocols: HTTP, SMTP, FTP, TFTP, Telnet, URL. Installation of text editors (Notepad++), use of editors, file creation, file editing, and file saving.	
Unit-II	15
HTML: Introduction, Basic Structure of HTML, Head Section and Elements of Head Section, Formatting Tags: Bold, Italic, Underline, Strikethrough, Div, Pre Tag. Links, Hyperlink and Hypermedia. Tables, Table Attributes, Cellpadding, Cellspacing, Rowspan and Colspan. Lists and their types. Forms, Form Elements, Frames and Framesets.	
Unit-III	15
Java Script: Overview of JavaScript, Object orientation and JavaScript, General syntactic characteristics, Primitives, Operations and expressions, Screen output and keyboard input; Control statements, Object creation and Modification, Arrays, Functions, Pattern matching using expressions, Errors in scripts.	
Unit-IV	15
CSS: Introduction to CSS, Types of CSS, CSS Selectors: Universal Selector, ID selector, Tag Selector, Class Selector, Grouping Selector, Attribute Selector. CSS Colors and Backgrounds, CSS Fonts and Text, Box Model, List properties, Border Properties, Positioning Properties, CSS Lists, CSS Tables. CSS Transitions, Transformations, Animations.	

References:

1. Web Development and Design Foundations with HTML5, Terry Felke-Morris, Pearson, 10th Edition (2023).
2. Learning Web Design, Jennifer Robbins, O'Reilly Media, 6th Edition (2024)
3. JavaScript and jQuery: Interactive Front-End Web Development, Jon Duckett, Wiley.
4. HTML5 and CSS3 All-in-One For Dummies, Andy Harris, Wiley, 4th Edition (2022).

CA3T2: Data Mining

Course Code: CA5T2	Course Title: Data Mining
Course Credits: 04	Teaching Hours/Week: 04
Total Contact Hours: 60	Exam Duration : 3 Hours
Internal Assessment : 20 Marks	Semester End Exam: 80 Marks

Course Outcomes (COs): After the successful completion of the course, the student will be able to:

CO1: Explain the fundamental concepts of data mining, data warehousing, ETL processes, knowledge discovery, and various data mining tasks and techniques.

CO2: Apply statistical and decision tree-based classification algorithms to solve classification problems.

CO3: Analyze clustering techniques using similarity and distance measures, hierarchical and partitioning algorithms, and identify outliers in datasets.

CO4: Apply association rule mining techniques, including Apriori, Sampling, Partitioning, and Parallel algorithms, to discover meaningful patterns and relationships in large datasets.

Course Content	Hours
Unit-I	15
Basic Data Mining Tasks: Introduction to data warehousing, ETL Tools, Classification, Regression, Time Series Analysis, Prediction, Clustering, Summarization, Association Rules, Sequence Discovery. Data Mining Concepts: Data Mining versus Knowledge Discovery in Databases, The Development of Data Mining, Data Mining Issues, Data Mining Metrics, Social Implications of Data Mining, Data Mining from a Database Perspective. Data Mining Techniques: Statistical Perspective on Data Mining, Similarity Measures, Decision Trees.	
Unit-II	15
Classification: Introduction, Statistical-Based Algorithms: Regression, Bayesian Classification, Distance-Based Algorithms: Simple Approach, K-Nearest Neighbors, Decision Tree-Based Algorithms: ID3, C4.5, CART, Scalable Decision Tree Techniques.	
Unit-III	15
Clustering: Introduction, Similarity and Distance Measures, Outliers, Hierarchical Algorithms: Agglomerative Algorithms, Divisive Clustering, Partitional Algorithms: Minimum Spanning Tree, Squared Error Clustering Algorithm, K-Means Clustering, Nearest Neighbor Algorithm.	
Unit-IV	15
Association Rules: Introduction, Large Itemsets, Basic Algorithms, Apriori Algorithm, Sampling Algorithm, Partitioning, Parallel and Distributed Algorithms: Data Parallelism, Task Parallelism, Comparing Approaches, Incremental Rules.	

References:

1. Introduction to Data Mining, Pang-Ning Tan, Michael Steinbach, Vipin Kumar, Addison-Wesley, Second Edition, 2018.
2. Data Mining: Concepts and Techniques (3rd Edition), Data Mining: Concepts and Techniques by Jiawei Han, Micheline Kamber, and Jian Pei, Morgan Kaufmann.
3. Introduction to Data Mining, Pang-Ning Tan, Michael Steinbach, Vipin Kumar, Pearson/Addison-Wesley.
4. Data Warehousing Fundamentals, Paulraj Ponniah, Wiley.

CA3T3: Software Engineering

Course Code: CA3T3	Course Title: Software Engineering
Course Credits: 04	Teaching Hours/Week: 04
Total Contact Hours: 60	Exam Duration : 3 Hours
Internal Assessment : 20 Marks	Semester End Exam: 80 Marks

Course Outcomes (COs): After the successful completion of the course, the student will be able to:

CO1: Explain the fundamentals of software engineering, software process models, and agile software development methodologies used in modern software projects.

CO2: Analyze software requirements, perform requirements engineering activities, identify project risks, and apply estimation and project planning techniques for software development.

CO3: Apply software design principles, UML modeling techniques, architectural design concepts, and software testing strategies to develop reliable software systems.

CO4: Evaluate software quality assurance practices, software reliability measures, review techniques, and release management strategies to ensure high-quality software delivery.

Course Content	Hours
Unit-I	15
Introduction: The evolving role of software, changing nature of software, layered technology, a process framework. Process models: The waterfall model, incremental process models, evolutionary process models, the unified process. Agile software development: Agility Principles, Agile methods, Plan-driven and agile development, Extreme programming, Scrum, A Tool Set for the Agile Process.	
Unit-II	15
Software Requirements Engineering: Functional and non-functional requirements, The software requirements document, Requirements specification, Requirements engineering processes, Requirements elicitation and analysis, Requirements validation, Requirements management. Risk Management: Reactive Vs proactive risk strategies, Software risks, Risk identification, Risk projection, Risk refinement, RMMM, RMMM plan, Project Planning: Software pricing, Plan-driven development, Project scheduling, Agile planning, Estimation techniques.	
Unit-III	15
Design: Design process and design quality, Design concepts, The design model, Software architecture, Data design, Architectural design, Basic structural modeling, Class diagrams, Sequence diagrams, Collaboration diagrams, Use case diagrams, Component diagrams, Testing Strategies: A strategic approach to software testing, Test strategies for conventional software, Black-box and white-box testing, Validation testing, System testing.	
Unit-IV	15
Quality Management: Quality concepts, Software quality assurance, Software reviews, Formal technical reviews, Statistical software quality assurance, Software reliability. Release Management: Release planning, Development and build plans, Release strategies, Risk management, Post-deployment monitoring.	

References:

1. Roger S. Pressman and Bruce R. Maxim, Software Engineering: A Practitioner's Approach, 9th Edition, McGraw-Hill Education, 2020.
2. Software Engineering, Ian Sommerville, 10th Edition, Pearson Education, 2016.
3. Pankaj Jalote, Software Engineering: A Precise Approach, Wiley India, 2010.
4. Software Engineering, a book by Aggarwal K.K, Singh Yogesh, New Age International

CA5P1: Web Technologies Lab

Course Code: CA5P1	Course Title: Web Technologies Lab
Course Credits: 02	Teaching Hours/Week: 04
Total Contact Hours: 60	Exam Duration : 3 Hours
Internal Assessment : 10 Marks	Semester End Exam: 40 Marks

Course Outcomes (COs): After the successful completion of the course, the student will be able to:

CO1: Develop web pages using HTML text formatting, lists and tables by applying standard HTML tags and attributes.

CO2: Develop JavaScript programs to perform arithmetic operations, evaluate mathematical expressions, and implement user-defined functions for solving computational problems.

CO3: Demonstrate the use of internal, external and inline style sheets.

CO4: Develop web pages implementing the CSS Box Model.

Section - A

1. Write a HTML program by using text formatting tags.
2. Write a HTML program using presentational element tags `<b>`, `<i>`, `<strike>`, `<sup>`, `<sub>`, `<big>`, `<small>`, `<hr>`.
3. Write a program in create a HTML page, which has properly aligned paragraphs with image along with it.
4. Write a HTML program to demonstrate the following:
(i) Unordered List (ii) Ordered List (iii) Definition List.
5. Create a table using HTML to display your class timetable. Use your choice of background color, table color etc. Use appropriate cell spacing, cell padding and cell width.
6. Create a form having number of elements (Textboxes, Radio buttons, Checkboxes, and so on). Write JavaScript code to count the number of elements in a form.
7. Write a JavaScript program to perform arithmetic operations using user Input.
8. Write a JavaScript program to find factorial of a given number using User defined function.
9. Develop a HTML Form, which accepts any Mathematical expression. Write JavaScript code to Evaluates the expression and Displays the result.
10. Create a HTML form that has number of Textboxes. When the form runs in the Browser fill the textboxes with data. Write JavaScript code that verifies that all textboxes has been filled. If a textboxes has been left empty, popup an alert indicating which textbox has been left empty.

Section - B

1. Create a form for Student information. Write JavaScript code to find Total, Average, Result and Grade.
2. Create a form for Employee information. Write JavaScript code to find DA, HRA, PF, TAX, Gross pay, Deduction and Net pay.
3. Write a HTML program to create login form and verify username and password.
4. Write a JavaScript program to calculate area of rectangle using function.
5. Write a program using Cascading Style Sheets (CSS) to display image on your choice and write few lines about it.
6. Create an HTML document that demonstrates the use of an internal style sheet in CSS
7. Create an HTML document that demonstrates the use of an external style sheet in CSS
8. Create an HTML document that demonstrates the use of an inline style sheet in CSS
9. Develop a CSS program to demonstrate the placement of images at different positions on a web page.
10. Develop a CSS program to demonstrate the implementation of the CSS Box Model.

CA5P2: Data Mining Lab

Course Code: CA5P2	Course Title: Data Mining Lab
Course Credits: 02	Teaching Hours/Week: 04
Total Contact Hours: 60	Exam Duration : 3 Hours
Internal Assessment : 10 Marks	Semester End Exam: 40 Marks

Note: Students will be expected to write and execute data mining lab programs using Python.

Course Outcomes (COs): After the successful completion of the course, the student will be able to:

CO1: Understand and implement fundamental data preprocessing techniques, including Extract, Transform and Load (ETL), data cleaning, and data transformation using Python.

CO2: Implement basic machine learning techniques such as K-Nearest Neighbours (KNN) classification and outlier detection methods for data analysis.

CO3: Apply the Knowledge Discovery in Databases (KDD) process to perform data selection, preprocessing, transformation, and prepare datasets for analysis using Python.

CO4: Develop and implement Simple Linear Regression models to predict continuous values and interpret the prediction results using Python libraries.

Section-A

1. Write a Python program to perform the Extract, Transform and Load (ETL) process on sales data by converting the sales values into integers and displaying the transformed data.

Data: Sales = ["1200", "1500", "1800", "2100", "2500"]

2. Write a Python program to calculate the mean, median and mode of the given student marks.

Data: Marks = [65, 72, 80, 72, 90, 68, 75]

3. Write a Python program to calculate the Pearson correlation coefficient between study hours and marks obtained.

Data: Study Hours = [2, 3, 4, 5, 6, 7]

Marks = [40, 50, 60, 70, 80, 90]

4. Write a Python program to calculate the Euclidean distance between two data objects.

Data: Customer A = [25, 45000]

Customer B = [30, 52000]

5. Write a Python program to classify a customer using the K-Nearest Neighbours (KNN) algorithm.

Data: Income = [[20], [25], [35], [45], [55], [65]]

Class = ["Low", "Low", "Medium", "Medium", "High", "High"]

6. Write a Python program to identify outliers using the Interquartile Range (IQR) method.

Data: Data = [12, 14, 15, 16, 17, 18, 19, 22, 95]

7. Write a Python program to calculate the support value of the given itemset from the transaction database.

Data:

```
Transactions = [
    ["Milk", "Bread"],
    ["Milk", "Eggs"],
    ["Milk", "Bread", "Butter"],
    ["Bread", "Butter"],
    ["Milk", "Bread"]
]
```

8. Write a Python program to calculate the confidence of an association rule.

Data:

```
Transactions = [  
    ["Milk", "Bread"],  
    ["Milk", "Bread", "Butter"],  
    ["Milk"],  
    ["Bread"],  
    ["Milk", "Bread"]  
]  
Rule = Milk → Bread
```

9. Write a Python program to calculate the Manhattan distance between two data points.

10. Write a Python program to calculate the Cosine similarity between two vectors.

Section-B

1. Write a Python program to implement the Knowledge Discovery in Databases (KDD) process by performing data selection, preprocessing and transformation before displaying the final dataset.

Data:

Name	Age	Salary
Anil	24	35000
Bharath	26	42000
Charan	25	-
Deepa	28	50000
Farah	27	48000

2. Write a Python program to predict house prices using Simple Linear Regression and display the predicted price for a house of 2000 sq.ft.

Data:

```
House Size = [1200, 1500, 1800, 2100, 2400]  
Price = [48, 60, 72, 84, 96]
```

3. Write a Python program to classify students using the Gaussian Naive Bayes algorithm and display the prediction and accuracy.

Data:

```
Hours = [[2], [3], [4], [5], [6], [7]]  
Result = ["Fail", "Fail", "Pass", "Pass", "Pass", "Pass"]
```

4. Write a Python program to classify loan applications using the Decision Tree algorithm and display the generated decision tree.

Data:

```
Income = [25000, 30000, 35000, 50000, 60000, 70000]  
Age = [22, 25, 30, 35, 40, 45]  
Loan Status = ["Reject", "Reject", "Reject", "Approve", "Approve", "Approve"]
```

5. Write a Python program to perform customer segmentation using the K-Means clustering algorithm and display the cluster assigned to each customer.

Data:

```
Customer Age = [22, 25, 28, 40, 42, 45]  
Spending Score = [20, 25, 30, 70, 75, 80]
```

6. Write a Python program to perform Agglomerative Hierarchical Clustering and display the clusters formed.

7. Write a Python program to perform Divisive Hierarchical Clustering on the given dataset.

8. Write a Python program to implement the Apriori algorithm and generate frequent itemsets from the given transaction database.

Data:

```
Transactions = [
    ["Milk","Bread","Eggs"],
    ["Milk","Bread"],
    ["Milk","Eggs"],
    ["Bread","Butter"],
    ["Milk","Bread","Butter"]]
```

9. Write a Python program to demonstrate Data Parallelism by dividing a list of numbers into two equal parts and processing them separately.

10. Write a Python program to compare Sequential Processing and Parallel Processing by calculating the execution time required to find the sum of a large list of numbers.

SEC2: Data Analytics Lab

Course Code: SEC2	Course Title: Data Analytics Lab
Course Credits: 02	Teaching Hours/Week: 04
Total Contact Hours: 60	Exam Duration : 3 Hours
Internal Assessment : 10 Marks	Semester End Exam: 40 Marks

Course Outcomes (COs): After the successful completion of the course, the student will be able to:

CO1: Apply fundamental spreadsheet functions and formatting techniques, including Conditional Formatting.

CO2: Analyze datasets using advanced spreadsheet features such as Pivot Tables, Data Validation, etc.

CO3: Develop Python programs to perform one-sample, two-independent-samples, and paired t-tests for hypothesis testing.

CO4: Develop Python programs to perform statistical significance tests using one-way ANOVA and the Chi-Square test and the correlation coefficient.

Note: 1. Practical batches, Internal Assessment (IA) scheme and Semester End Exam (SEE) to be followed as per regular practical subject.

2. Download sample datasets from open sources (Kaggle.com, etc.) to practice these functions.

Section – A Spreadsheet (Excel)

1. Demonstrate the use of Conditional Formatting, IF, COUNTIF, SUMIF, AVERAGE and CONCAT functions.
2. Demonstrate the use of INDEX, MATCH, UNIQUE, IFS, COUNTIFS, SUMIFS and AVERAGEIFS functions.
3. Demonstrate the use of VLOOKUP, HLOOKUP, XLOOKUP, COUNT and COUNTA functions.
4. Demonstrate the use of LEFT, MID, RIGHT, LEN, SUBSTITUTE, SEARCH and ISNUMBER functions.
5. Demonstrate the use of TODAY, NOW, YEAR, MONTH, NETWORKDAYS and EOMONTH functions.
6. Demonstrate the use of OFFSET, CHOOSE, LET, MAX, SORT, SORTBY and RANK functions. .
7. Demonstrate the use of FILTER, FREQUENCY, SEQUENCE, RANDARRAY and IFERROR functions.
8. Demonstrate the use of Pivot Tables, What-If Analysis, Data Validation, Subtotals with Ranges.

Section – B

1. Develop a Python program to calculate the probability of simple events based on user input.
2. Develop a Python program to analyze an exam pass/fail prediction scenario using a probability distribution.
3. Develop a Python program to perform a one-sample t-test.
4. Develop a Python program to perform a two-independent-samples t-test.
5. Develop a Python program to perform a paired t-test.
6. Develop a Python program to perform a test of significance using one-way ANOVA.
7. Develop a Python program to perform a test of significance using Chi-Square Test.
8. Develop a Python program to plot a scatter diagram and compute the correlation coefficient.

CA5P3: Internship

Course Code: CA5P3	Course Title: Internship
Course Credits: 02	Teaching Hours/Week: 04
Total Contact Hours: 60	Exam Duration : 3 Hours
Internal Assessment : 10 Marks	Semester End Exam: 40 Marks

General Guidelines:

1. The primary goal of an internship is to gain practical work experience, often required as part of the curriculum for the BCA program. Internships are typically completed during the Fifth semester, though they may also be undertaken at the end of the Fourth semester during vacation periods.
2. The internship shall be undertaken on a part-time basis during the semester and must comprise 90 hours, to be completed over a period of 8 to 12 weeks.
3. Students shall undertake the internship either individually or in groups of up to three students.
4. It is recommended that the Internship report be approximately 40–60 pages long, typed on A4-size paper using Times New Roman font, with 1.5-line spacing and proper margins as follows:

FORMAT OF INTERNSHIP REPORT

1. COVER PAGE
2. CERTIFICATE ISSUED BY THE COMPANY
3. COLLEGE CERTIFICATE
4. STUDENT DECLARATION
5. ACKNOWLEDGEMENT
6. TABLE OF CONTENTS
7. EXECUTIVE SUMMARY
8. CHAPTER I: INTRODUCTION
9. CHAPTER II: DESCRIPTION OF THE ORGANIZATION
10. CHAPTER III: EXPERIENTIAL LEARNING
11. CHAPTER IV: INTERNSHIP OUTCOMES AND CONCLUSION
12. BIBLIOGRAPHY

Semester: VI

CA6T1: PHP and MySQL

Course Code: CA6T1	Course Title: PHP and MySQL
Course Credits: 04	Teaching Hours/Week: 04
Total Contact Hours: 60	Exam Duration : 3 Hours
Internal Assessment : 20 Marks	Semester End Exam: 80 Marks

Course Outcomes (COs): After the successful completion of the course, the student will be able to:

CO1: Explain the fundamentals of PHP, including its syntax, data types, variables, constants, operators, expressions, and implement decision-making and looping constructs to develop basic PHP programs.

CO2: Develop PHP programs using arrays, functions, file inclusion techniques, type casting, and date/time functions to solve computational problems effectively.

CO3: Design PHP applications using strings, object-oriented programming concepts and form handling techniques for dynamic web applications.

CO4: Build database-driven web applications by integrating PHP with MySQL to perform database connectivity and execute operations efficiently.

Course Content	Hours
Unit-I	15
Introduction to PHP: Introduction to PHP, History and Features of PHP, Installation & Configuration of PHP, Embedding PHP code in Your Web Pages, Understanding PHP, HTML and White Space, Writing Comments in PHP, Sending Data to the Web Browser, Data types in PHP, Keywords in PHP, Using Variables, Constants in PHP, Expressions in PHP, Operators in PHP. Programming with PHP: Conditional statements: if, if-else, switch, The ? Operator, Looping statements: while Loop, do-while Loop, for Loop	
Unit-II	15
Arrays in PHP: Introduction-Creating Arrays, Accessing Array elements, Types of Arrays: Indexed v/s Associative arrays, Multidimensional arrays, Creating Array, Accessing Array, Manipulating Arrays, Displaying array, Using Array Functions, Including and Requiring Files- use of Include() and Require(), Implicit and Explicit Casting in PHP. Using Functions , Class- Objects, Forms in PHP: Functions in PHP, Function definition, Creating and invoking user-defined functions, Formal parameters versus actual parameters, Function and variable scope, Recursion, Library functions, Date and Time Functions	
Unit-III	15
Strings in PHP: Creating and Declaring String, String Functions. Class & Objects in PHP: What is Class & Object, Creating and accessing a Class & Object, Object properties, object methods, Overloading, inheritance, Constructor and Destructor Form Handling: Creating HTML Form, Handling HTML Form data in PHP.	
Unit-IV	15
Database Handling Using PHP with MySQL: Introduction to MySQL: Database terms, Data Types. Accessing MySQL –Using MySQL Client and Using php MyAdmin, MySQL Commands, Using PHP with MySQL: PHP MySQL Functions, Connecting to MySQL and Selecting the Database, Executing Simple Queries, Retrieving Query Results, Counting Returned Records, Updating Records with PHP.	

References:

1. PHP and MySQL Web Development, Luke Welling & Laura Thomson, Addison-Wesley.
2. Learning PHP, MySQL & JavaScript, Robin Nixon, O'Reilly Media
3. PHP 8 and MySQL Web Development, Luke Welling & Laura Thomson, Addison-Wesley.
4. PHP: The Complete Reference, McGraw-Hill.

CA6T2: Mobile Application Development

Course Code: CA6T2	Course Title: Mobile Application Development
Course Credits: 04	Teaching Hours/Week: 04
Total Contact Hours: 60	Exam Duration : 3 Hours
Internal Assessment : 20 Marks	Semester End Exam: 80 Marks

Course Outcomes (COs): After the successful completion of the course, the student will be able to:

CO1: Explain the architecture, features, development framework, and tools of the Android operating system, and develop basic Android applications using Android Studio and Android Virtual Devices.

CO2: Design Android applications by implementing core application components and configuring the Android Manifest file with appropriate permissions and intent filters.

CO3: Develop interactive Android user interfaces using layouts, UI widgets, animations, application resources, and preferences, and perform testing and publishing of Android applications.

CO4: Build Android applications that utilize data storage, SQLite databases, Content Providers, networking, and web APIs, and deploy applications for real-world use.

Course Content	Hours
Unit-I	15
Android OS design and Features: Android development framework, SDK features, Installing and running applications on Android Studio, Creating AVDs, Types of Android applications, Best practices in Android programming, Android tools, Building your First Android application.	
Unit-II	15
Android Application Design Essentials: Anatomy of an Android applications, Android terminologies, Application Context, Activities, Services, Intents, Receiving and Broadcasting Intents, Android Manifest File and its common settings, Using Intent Filter, Permissions.	
Unit-III	15
Android User Interface Design Essentials: User Interface Screen elements, Designing User Interfaces with Layouts, Drawing and Working with Animation. Testing Android applications, Publishing Android application, Using Android preferences, Managing Application resources in a hierarchy, working with different types of resources.	
Unit-IV	15
Using Common Android APIs: Using Android Data and Storage APIs, Managing data using Sqlite, Sharing Data between Applications with Content Providers, Using Android Networking APIs, Using Android Web APIs, Deploying Android Application to the World.	

References:

1. Beginning Android Application Development, Wei-Meng Lee, Wiley Publications.
2. Android Application Development All-in-One For Dummies, Barry Burd, Wiley Publications.
3. Android Programming for Beginners, John Horton, Packt Publishing.
4. Beginning Android Programming with Android Studio, Jerome (J.F.) DiMarzio, Wiley Publications.

CA6T3: Deep Learning

Course Code: CA6T3	Course Title: Deep Learning
Course Credits: 04	Teaching Hours/Week: 04
Total Contact Hours: 60	Exam Duration : 3 Hours
Internal Assessment : 20 Marks	Semester End Exam: 80 Marks

Course Outcomes (COs): After the successful completion of the course, the student will be able to:

CO1: Explain the fundamental concepts of Deep Learning models, including Artificial Neural Networks, Perceptrons, Multilayer Perceptrons, Feedforward Networks, and the Backpropagation learning algorithm.

CO2: Analyze the architecture and working principles of Convolutional Neural Networks, activation functions, pooling techniques, and modern CNN architectures for computer vision applications.

CO3: Apply Recurrent Neural Networks, Long Short-Term Memory and Gated Recurrent Units to solve sequential data analysis problems while addressing challenges such as the vanishing gradient problem.

CO4: Explain the concepts, architecture, training process, and applications of Generative Adversarial Networks and compare discriminative and generative learning models.

Course Content	Hours
Unit-I	15
Introduction to Deep Learning: Historical Trends in Deep Learning, Why Deep Learning Is Growing, Artificial Neural Networks (ANN), Nonlinear Classification Using Neural Networks: XOR/XNOR, Single-Layer Perceptron, Multilayer Perceptron, Feedforward Network, Deep Feedforward Networks, Stochastic Gradient-Based Learning, Hidden Units, Architecture Design, and Backpropagation.	
Unit-II	15
Convolutional Neural Networks (CNNs): Introduction to CNNs and their applications in computer vision; basic CNN architecture; activation functions (Sigmoid, Tanh, ReLU, and Softmax); types of pooling layers; training CNNs using TensorFlow; popular CNN architectures such as VGG, GoogLeNet, and ResNet; dropout, normalization, and data augmentation.	
Unit-III	15
Recurrent Neural Networks (RNNs): Introduction to Recurrent Neural Networks (RNNs) and their applications in sequential data analysis; Backpropagation Through Time (BPTT); the vanishing gradient problem and gradient clipping; Long Short-Term Memory (LSTM) networks; Gated Recurrent Units (GRUs); Bidirectional LSTM networks; and Bidirectional Recurrent Neural Networks (BiRNNs).	
Unit-IV	15
Generative Adversarial Networks (GANs): Introduction to generative models, concepts and principles of GANs, architecture of GANs (generator and discriminator networks), comparison between discriminative and generative models, training of Generative Adversarial Networks (GANs), and applications of GANs.	

References:

1. Simon J.D. Prince, Understanding Deep Learning, MIT Press, 2023.
2. Josh Patterson and Adam Gibson, Deep Learning: A Practitioner's Approach, 2nd Edition, O'Reilly Media, 2021.
3. François Chollet, Deep Learning with Python, 2nd Edition, Manning Publications, 2021.
4. Aston Zhang, Zachary C. Lipton, Mu Li, and Alexander J. Smola, Dive into Deep Learning, Cambridge University Press, 2023.

CA6P1: PHP and MySQL Lab

Course Code: CA6P1	Course Title: PHP and MySQL Lab
Course Credits: 02	Teaching Hours/Week: 04
Total Contact Hours: 60	Exam Duration : 3 Hours
Internal Assessment : 10 Marks	Semester End Exam: 40 Marks

Course Outcomes (COs): After the successful completion of the course, the student will be able to:

CO1: Apply PHP syntax, operators, control structures, functions, and string/array manipulation techniques to develop basic server-side programs.

CO2: Develop PHP applications using form handling techniques (GET and POST methods), recursive functions, and built-in date/time functions to process user input.

CO3: Design and implement PHP programs for file handling, file uploading, and object-oriented programming concepts such as constructors and destructors.

CO4: Create and manage MySQL databases and tables, and integrate PHP with MySQL to perform database-driven web application development.

Section-A

1. Write a PHP script to find odd or even number from given number.
2. Write a PHP script to find maximum of three numbers.
3. Write a PHP script to swap two numbers.
4. Write a PHP script to find the factorial of a number.
5. Write a PHP script to check whether given number is palindrome or not..
6. Write a PHP script to reverse a given number and calculate its sum.
7. Write a PHP script to generate a Fibonacci series using Recursive function.
8. Write a PHP script to implement atleast seven string functions.
9. Write a PHP program to insert new item in array on any position in PHP.
10. Write a PHP script to implement constructor and destructor.

Section-B

1. Write a PHP script to implement form handling using get method.
2. Write a PHP script to implement form handling using post method.
3. Write a PHP script that receive form input by the method post to check the number is prime or not
4. Write a PHP script that receives string as a form input.
5. Write a PHP script to compute addition of two matrices as a form input.
6. Write a PHP script to show the functionality of date and time function.
7. Write a PHP program to upload a file.
8. Write a PHP script to implement database creation.
9. Write a PHP script to create table.
10. Develop a PHP program to design a college admission form using MYSQL database.

CA6P2: Mobile Application Development Lab

Course Code: CA6P2	Course Title: Mobile Application Development Lab
Course Credits: 02	Teaching Hours/Week: 04
Total Contact Hours: 60	Exam Duration : 3 Hours
Internal Assessment : 10 Marks	Semester End Exam: 40 Marks

Course Outcomes (COs): After the successful completion of the course, the student will be able to:

CO1: Develop simple Android applications using basic UI components, event handling, layout managers, fonts, colors, menus, dialog boxes, and graphical primitives.

CO2: Build interactive Android applications, including calculators and other utility applications, by integrating widgets, event listeners, and application logic

CO3: Design and implement an Android application for sending e-mails using Intent and e-mail clients.

CO4: Design and implement a secure login application using SQLite database for user authentication.

Note: Download and install the open-source software Android Studio to practice these programs.

Section – A

1. Develop an application for Simple Counter.
2. Develop an application that uses GUI components, fonts and colours.
3. Develop an application to display your personal details using GUI Components.
4. Develop a Simple Calculator that uses radio buttons and text view.
5. Develop an application that uses Intent and Activity.
6. Develop an application that uses Dialog Boxes.
7. Develop an application to display a Splash Screen.
8. Develop an application that uses Layout Managers and event listeners.
9. Write an application that draws basic graphical primitives on the screen.
10. Develop an application that uses different types of Menus.

Section – B

1. Develop an application that uses to send messages from one mobile to another mobile.
2. Develop an application that uses to send E mail.
3. Develop a native application that uses GPS location information.
4. To develop a Android Application that that makes use of Notification Manager.
5. Develop an application that plays Audio and Video.
6. Develop an application that uses Local File Storage.
7. Develop an application for Simple Animation.
8. Develop an application that makes use of databases.
9. Develop an application for Login Page using Sqlite.
10. Develop an application for Student Mark sheet processing using Sqlite.

DSS1: Cyber Security

Course Code: DSS1	Course Title: Cyber Security
Course Credits: 02	Teaching Hours/Week: 02
Total Contact Hours: 30	Exam Duration : 1 ½ Hours
Internal Assessment : 10 Marks	Semester End Exam: 40 Marks

Course Outcomes (COs): After the successful completion of the course, the student will be able to:

CO1: Explain the fundamentals of cyberspace, internet technologies, cyber security concepts, and the architecture and governance of cyberspace.

CO2: Identify various types of cybercrimes, cyber threats, legal provisions under the IT Act, and the roles of organizations involved in cyber security and cybercrime prevention in India.

CO3: Examine the features, opportunities, risks, and security concerns associated with social media platforms, including relevant legal and ethical issues.

Course Content	Hours
Unit-I	10
Introduction to Cyber Security: Defining Cyberspace, Overview of Computer and Web Technology, Architecture of Cyberspace, Communication and Web Technology, Internet,	

World Wide Web, Advent of Internet, Internet Infrastructure for Data Transfer and Governance, Internet Society, Regulation of Cyberspace, Concept of Cyber Security, Issues and Challenges of Cyber Security.

Unit-II	10
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Cyber-crime and Cyber law: Classification of cybercrimes, Common cyber-crimes, cyber-crime targeting computers and mobiles, cyber crime against women and children, financial frauds, social engineering attacks, malware and ransomware attacks, zero day and zero click attacks, Cybercriminals modus-operandi, Reporting of cyber crimes, Remedial and mitigation measures, Legal perspective of cyber crime, IT Act 2000 and its amendments, Cyber-crime and offences, Organisations dealing with Cyber-crime and Cyber security in India, Case studies.

Unit-III	10
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Social Media Overview and Security: Introduction to Social networks, Types of Social media, Social media platforms, Social media monitoring, Hashtag, Viral content, Social media marketing, Social media privacy, Challenges, opportunities and pitfalls in online social network, Security issues related to social media, Flagging and reporting of inappropriate content, Laws regarding posting of inappropriate content, Best practices for the use of Social media, Case studies.

References:

1. Cyber Security: Understanding Cyber Crimes, Nina Godbole & Sunit Belapure, Wiley India.
2. Cyber Security and Cyber Laws, Alok Arora, BPB Publications.
3. Information Security: Principles and Practice, Mark Stamp, Wiley
4. Cyber Security, Nina Godbole, Wiley India

DSS1: Information Security

Course Code: DSS1	Course Title: Data and Information Security
Course Credits: 02	Teaching Hours/Week: 02
Total Contact Hours: 30	Exam Duration : 1 ½ Hours
Internal Assessment : 10 Marks	Semester End Exam: 40 Marks

Course Outcomes (COs): After the successful completion of the course, the student will be able to:

CO1: Explain the fundamental concepts of information security, including security terminology, historical developments, the need for security, and legal, ethical, and professional issues.

CO2: Analyze security structures, assess security requirements, and apply risk management principles, security planning, and government security standards in organizational environments.

CO3: Analyze the operation of classical and modern cryptographic algorithms and evaluate secure system design concepts, including trustworthy software, covert channels, inference, and single points of failure.

Course Content	Hours
Unit-I	10
Introduction, Definition of security, Assessing security, Security terminology , Historical developments, Structure of security, Introduction to Information Security, The Need for Security , Legal, Ethical, and Professional Issues in Information Security	
Unit-II	10
Risk Management and Special requirements such as Emanation Security/TEMPEST Standards, Planning for Security, Rainbow Series Reports for DOD; FIPS for all Federal Govt; DHS and CNSS guidance: Firewalls & VPNs	

Unit-III	10
Cryptography: Applications of cryptography, Terminology, Evolution of cryptography, Caesar ciphers, one-time pads, Operation of DES, AES, Public-key cryptosystems, Topics in Information Systems Security, Minimum privilege, Compartmentalization, Dual controls, Security perimeters, Trustworthy software, proof of design correctness, Single-points-of-failure, Covert channels, Inference	

References:

1. Principles of Information Security, Michael E. Whitman & Herbert J. Mattord, Cengage Learning.
2. Cryptography and Network Security: Principles and Practice, William Stallings, Pearson.
3. Information Security: Principles and Practice, Mark Stamp, Wiley.
4. Computer Security: Principles and Practice, William Stallings & Lawrie Brown, Pearson.

CA6P3: Project Work

Course Code: CA6P3	Course Title: Project Work
Course Credits: 04	Teaching Hours/Week: 08
Total Contact Hours: 60	Exam Duration : 3 Hours
Internal Assessment : 20 Marks	Semester End Exam: 80 Marks

General Guidelines:

1. Students shall complete the project work during the assigned laboratory hours. However, data collection may be undertaken outside the college and they must be implemented within the college.
2. The project must be domain-related, feasible, and include the phases of analysis, design, implementation, and testing. The work must also be original.
3. Students shall undertake the project either individually or in groups of up to three students.
4. It is recommended that the project report be approximately 60–100 pages long, typed on A4-size paper using Times New Roman font, with 1.5-line spacing and proper margins.
5. Internal Assessment Scheme for Project work.

Assessment Criteria	Marks
Attendance	5
Review / Presentation - I	5
Review / Presentation - II	5
Review / Presentation - III	5
Total	20

6. Evaluation Scheme for Project Work (SEE).

Assessment Criteria	Marks
Writing – Synopsis, Objectives, Data Flow Diagram, E R Diagram, Conclusion.	15
Project Demo and Presentation	40
Viva Voice	15
Project Report	10
Total	80